

Logging and Monitoring in BigQuery

08

Topics

01	Slot Projection and Cloud Monitoring
02	BigQuery Admin Panel
03	Cloud Audit Logs
04	Information Schema
05	Query Path and Common Errors



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01

Slot Projection and Cloud Monitoring

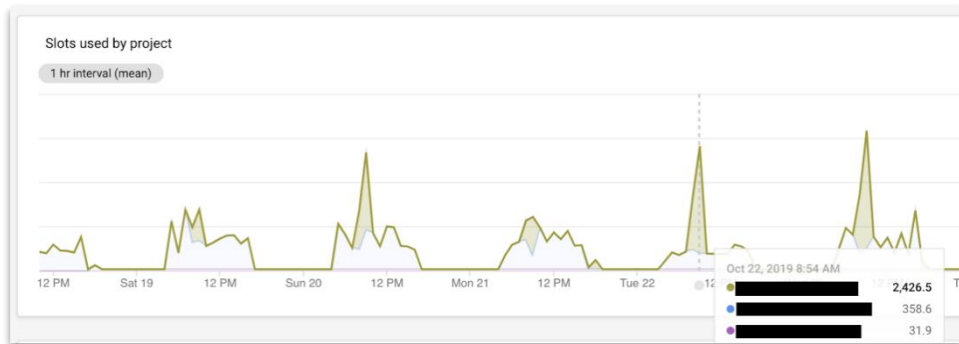
Slot Projection

Monitoring options



Cloud Monitoring

- Subset of [job metrics](#) available
- 375-project limit per workspace
- Limited reservations data



BQ Admin Panel

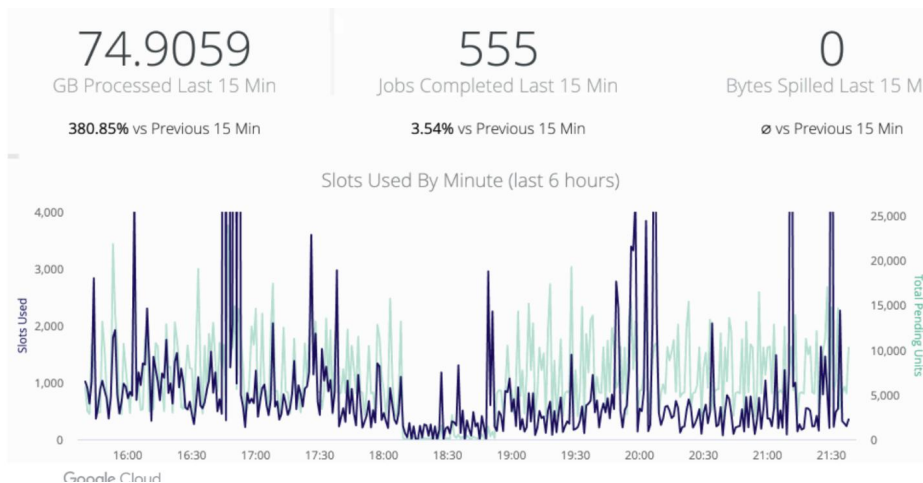
- Support [a list](#) of metrics types
- No alerting functionality
- Permissions required to view at org level



Recap | Slot projection: Look for the signals

How do you know whether you need more slots?

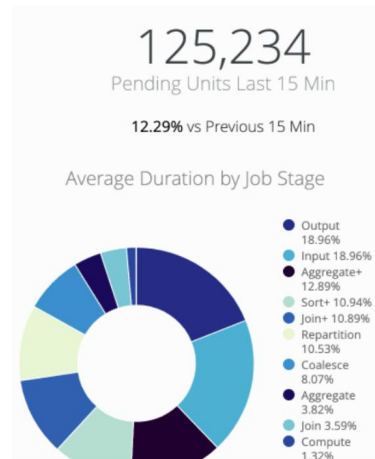
How do you know whether you actually need fewer slots?



- Monitor trend (week over week, month over month).
- Are more users being onboarded within same slot allocation?
 - Monitor user growth/trend.

Recap | Slot projection: Look for the signals

- More slots are needed for user queries if, for a given slot allocation, the following is observed:
 - Concurrency - consistently increasing
 - Throughput - consistently decreasing
 - Avg. Wait Time - consistently increasing
 - Avg. query run time - consistently increasing
 - Slot utilization - consistently increasing or remaining high; for example, 90% or higher
- Fewer slots needed if we see an opposite trend than above, and
 - An accurate company forecast shows a decreasing trend in usage. For example:
 - Fewer users
 - Fewer queries



Cloud Monitoring

Cloud Monitoring metrics

- What are metrics?
 - Metrics are numerical values that are collected at regular intervals and made available for analysis.

- How to use metrics?
 - Create charts and dashboards.
 - Trigger alerts for conditions or situations that need human intervention.
 - Analyze historical performance.



Cloud Monitoring

Metrics used in BigQuery

- Metrics in BigQuery:
 - The number of jobs that are running
 - How many bytes were scanned during a query
 - The distribution of query times
- Resource type in metrics:
 - `bigquery_dataset`
 - `bigquery_project`
 - `global`



Cloud Monitoring

(MQL) provides an expressive, text-based interface to Cloud Monitoring time-series data. By using MQL, you can retrieve, filter, and manipulate time-series data.

Google Cloud

qwiklabs-gcp-00-83f8195b0767

Search

Products, resources, docs (/)

Monitoring

Metrics Scope

1 project

Overview

Dashboards

Integrations

Services

Metrics explorer

Alerting

Uptime checks

Groups

Managed Prometheus

Permissions

Settings

Metrics explorer

EXPLORER

DIAGNOSTICS

CONFIGURATION

MQL

PROMQL

FORMAT

EXAMPLE QUERIES

Press Alt+F1 for Accessibility Options.

1

Line chart

1H

6H

1D

1W

10 sec interval

UTC-8

3:35 PM

3:40 PM

3:45 PM

3:50 PM

3:55 PM

4:00 P

Google Cloud

MQL example 1

Chart the number of interactive queries in flight

```
fetch global  
| metric 'bigquery.googleapis.com/query/count'  
| filter metric.priority = 'interactive'
```

MQL example 2

Get the number of load jobs in flight, grouped into 10-minute intervals

```
fetch bigquery_project  
| metric 'bigquery.googleapis.com/job/num_in_flight'  
| filter metric.job_type = 'load'  
| group_by 10m
```

Monitor BigQuery resources using Cloud Monitoring

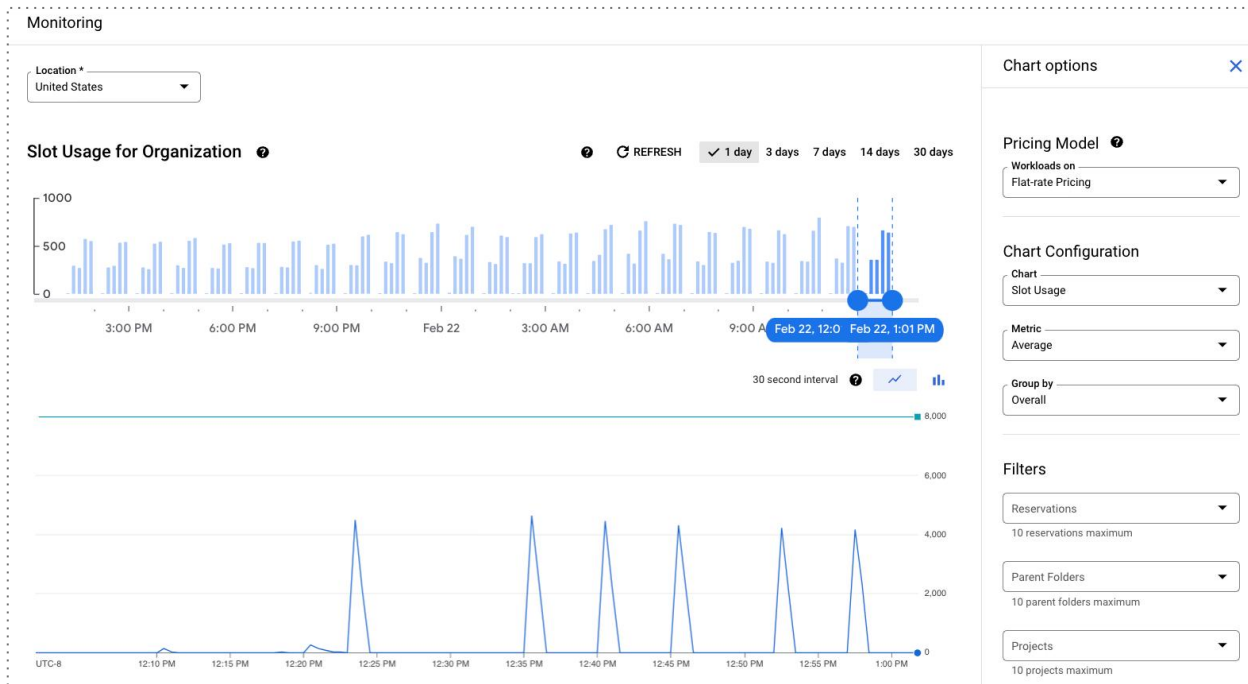
- Pre-requirement:
 - A Cloud Billing account
 - A BigQuery project with billing enabled
- How to monitor BigQuery resources in Cloud Monitoring:
 - View the Cloud Monitoring dashboard.
 - Visualize slots available and slots allocated.
 - Create a dashboard and chart.
 - Create an alert.



BigQuery Admin Panel

BigQuery admin panel

Proprietary + Confidential



Primary use cases:

Root-cause analysis for capacity planning based on historical resource utilization and diagnosing performance changes.

Data in the Admin Resource Charts is based on

and updates in real time, going back a maximum of 30 days.

Admin panel permissions

To view the full view in BigQuery Admin charts:

1. A **Google Cloud organization** is defined.
2. Have **one** of the following Identity and Access Management (IAM) permissions at the **organization level**:
 - ☐ `bigquery.jobs.listExecutionMetadata`
 - ☐ `bigquery.jobs.listAll`

An error message is displayed if you do not have the proper permissions assigned.

Permission denied. Please get `bigquery.jobs.listExecutionMetadata` or `bigquery.jobs.listAll` permission at organization level to access Admin Resource Charts.

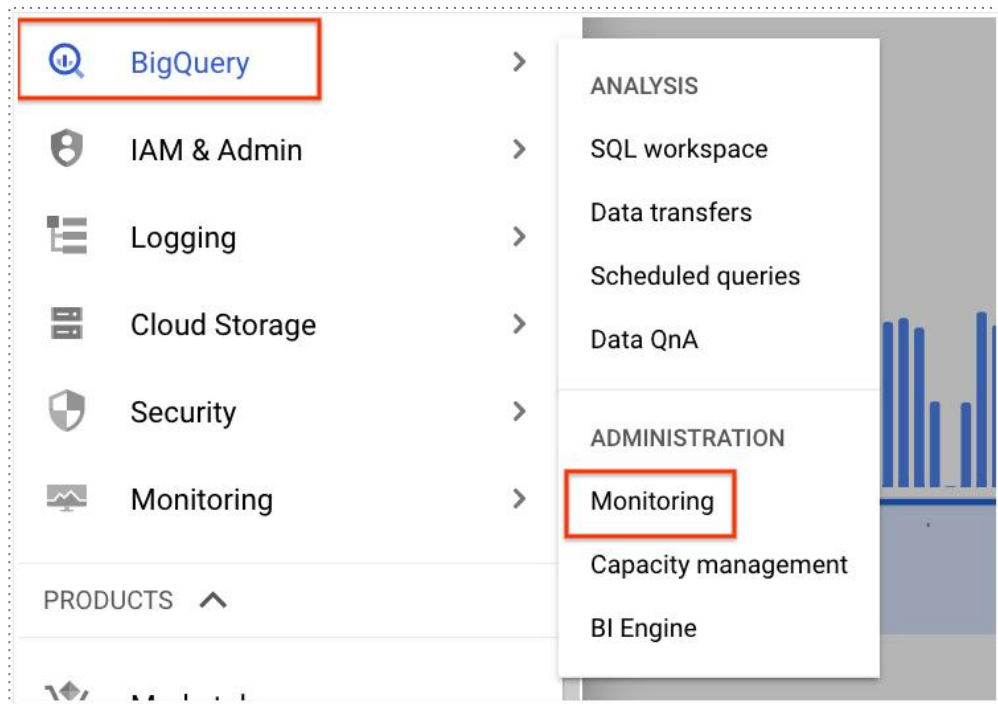


Admin panel permissions

- To view the partial data at the reservation level for reservation customers, you need `bigquery.reservations.list` and `bigquery.reservationAssignments.list`.
- To view the reservation capacity data on Admin Resource Charts, you need `bigquery.capacityCommitments.list`.
- You can toggle between on-demand pricing and flat-rate pricing in the Pricing Model section under Chart Options in the UI.

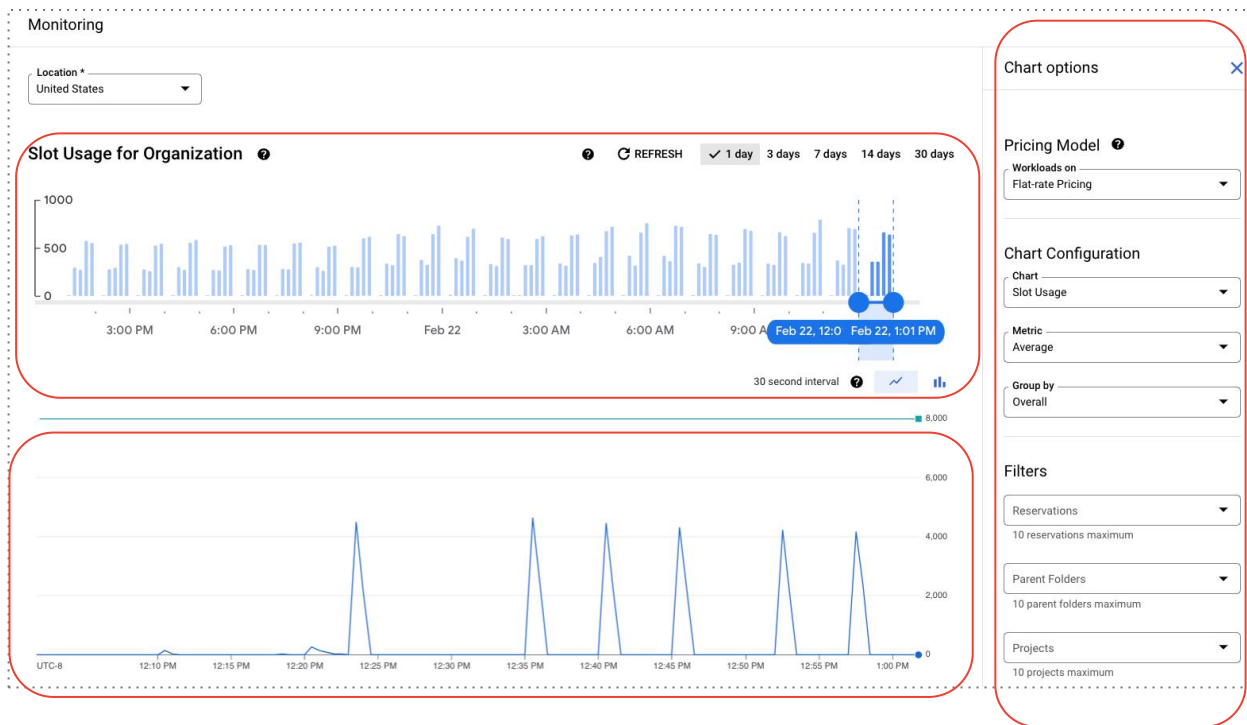


Using Admin Resource Charts



- In the project picker, select the project that you used to purchase slots and create reservations.
- On the BigQuery page, navigate to **ADMINISTRATION** > **Monitoring**.

Using Admin Resource Charts



Admin Resource Charts have three primary interactive components:

- The event timeline chart
- The main chart with different metrics:
 - Slot usage
 - Job concurrency
 - Job performance
 - Failed jobs
 - Bytes processed
- A side panel to pick different charts, breakdowns, and filter options

Admin panel billing

- Admin Resource Charts are available at no additional cost.
- Queries used to populate these charts are not billed and do not use slots in user-owned reservations.
- Queries that process too much data will time out.

03



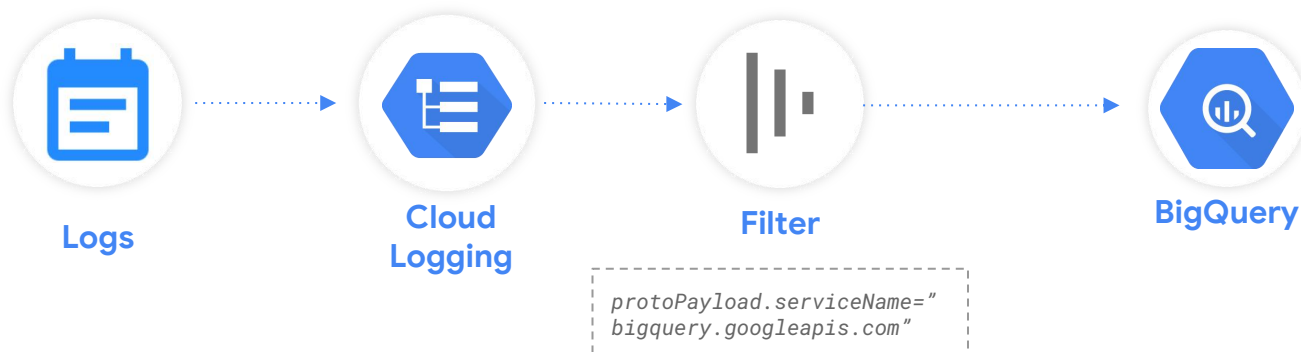
Cloud Audit Logs

BigQuery monitoring: Audit logs

Logs provide various details about the executed BigQuery job:

- **query** - The BigQuery SQL executed
- **startTime** - Time when the job started
- **endTime** - Time when the job ended
- **totalProcessedBytes** - Total bytes processed for a job
- **totalBilledBytes** - Processed bytes, adjusted by the job's CPU usage
- **totalSlotMs** - The total number of slot-ms consumed by the query job
- **referencedFields** - The columns of the underlying table that were accessed

Not included by default



Cloud Audit Logs can provide BigQuery-specific log information



BigQuery



Logging

BigQuery audit logs:

- Admin activity
- Data access
- System event

BigQuery provides three kinds of log messages

AuditData (older version)

The older AuditData is reported as the `protoPayload.serviceData` submessage within a LogEntry message.

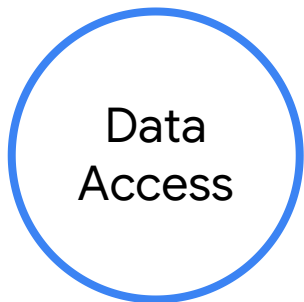
BigQueryAuditMetadata (newer version)

The newer **Use this for new work** BigQueryAuditMetadata is reported as the `protoPayload.metadata` within a LogEntry message. In these logs, the `protoPayload.serviceData` information is not set or used.

AuditLog

The logs that BigQuery Reservations and BigQuery Connections use when reporting requests.

Audit logs are organized into three log streams



Jobs:

`JobInsertion`
`JobChange`

Table data modifications:

`TableDataChange`
`TableDataRead`



You can set an expiration time on tables to remove them at a specified time.

A `TableDeletion` event is written to the system event stream when the table expires and is removed.

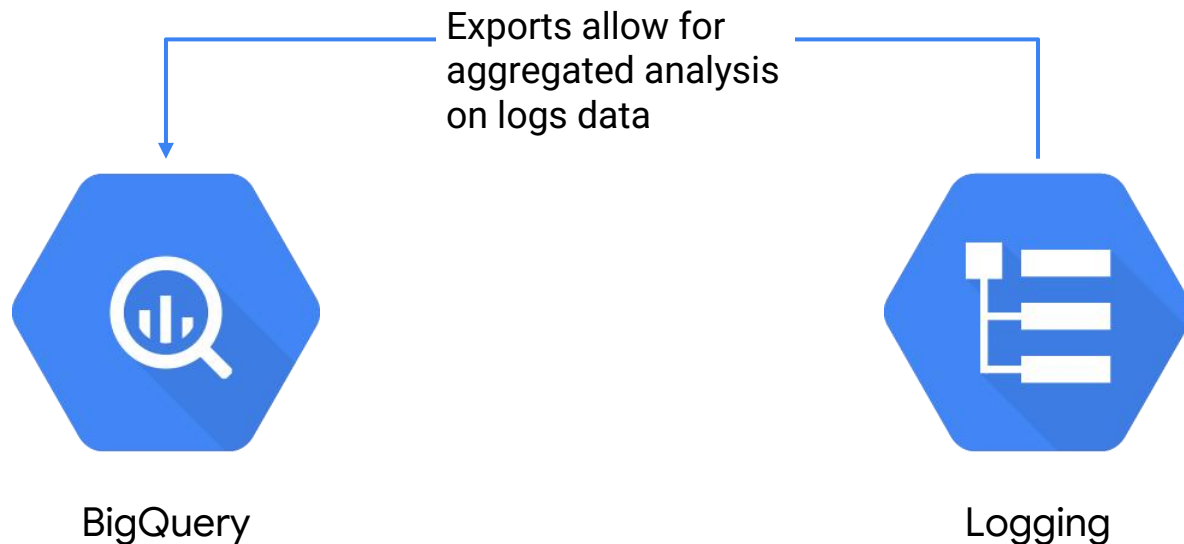


All remaining activities and events are reported as part of the main activity stream.

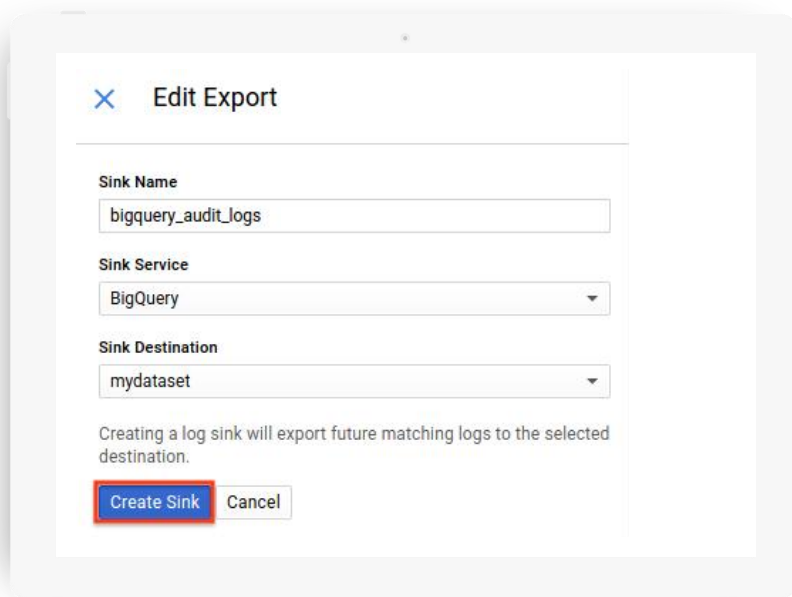
For example:

- Table creation
- Dataset creation

Log exports from Cloud Logging to BigQuery allows for aggregated analysis on log data



Best practice: Export Data Access audit logs to BigQuery



Edit Export

Sink Name
bigquery_audit_logs

Sink Service
BigQuery

Sink Destination
mydataset

Creating a log sink will export future matching logs to the selected destination.

Create Sink Cancel

Data access is exported in real time to BigQuery and includes metadata such as query executed, bytes scanned, and calling user.

Create **aggregated exports** to BigQuery at the organization or folder level to view data access across projects.

To filter relevant BigQuery audit messages, you can express filters as part of the export

```
protoPayload.metadata["@type"]="type.googleapis.com/google.cloud.audit.BigQueryAuditMetadata"
```

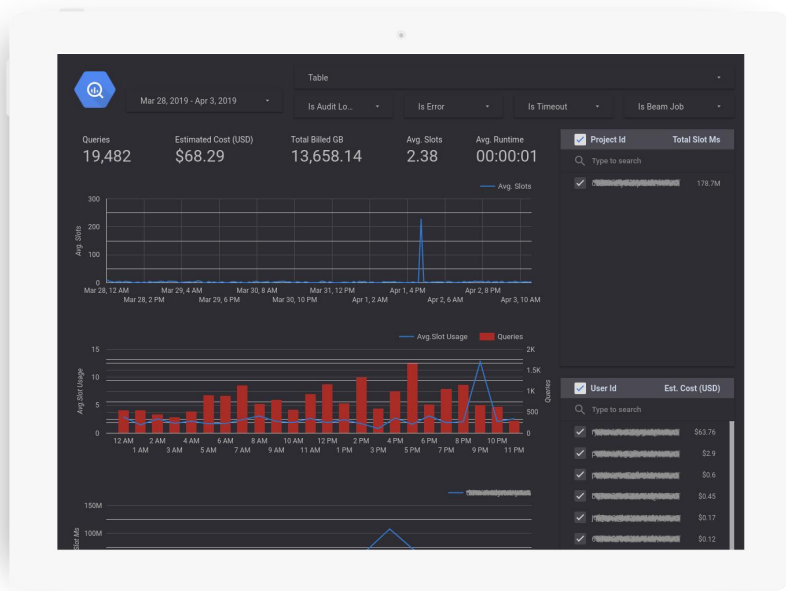
Querying exported logs: Report expired tables

```
#standardSQL
SELECT
  protopayload_auditlog.resourceName AS resourceName,
  receiveTimestamp as logTime
FROM
  `MYPROJECTID.MYDATASETID.cloudaudit_googleapis_com_system_event_201901*`
WHERE
  protopayload_auditlog.methodName = 'InternalTableExpired'
ORDER BY resourceName
```

Querying exported logs: Most popular datasets

```
#standardSQL
SELECT
  REGEXP_EXTRACT(protopayload_auditlog.resourceName, '^projects/[^/]+/datasets/([^/]+)/tables') AS datasetRef,
  COUNT(DISTINCT REGEXP_EXTRACT(protopayload_auditlog.resourceName,
    '^projects/[^/]+/datasets/[^/]+/tables/(.*)$')) AS active_tables,
  COUNTIF(JSON_EXTRACT(protopayload_auditlog.metadataJson, "$.tableDataRead") IS NOT NULL) AS dataReadEvents,
  COUNTIF(JSON_EXTRACT(protopayload_auditlog.metadataJson, "$.tableDataChange") IS NOT NULL) AS
dataChangeEventEvents
FROM `MYPROJECTID.MYDATASETID.cloudaudit_googleapis_com_data_access_2019*`
WHERE
  JSON_EXTRACT(protopayload_auditlog.metadataJson, "$.tableDataRead") IS NOT NULL
  OR JSON_EXTRACT(protopayload_auditlog.metadataJson, "$.tableDataChange") IS NOT NULL
GROUP BY datasetRef
ORDER BY dataReadEvents DESC
```

Best practice: Visualize audit logs



[github project](#)

Analyze spending trends and query volume over time.

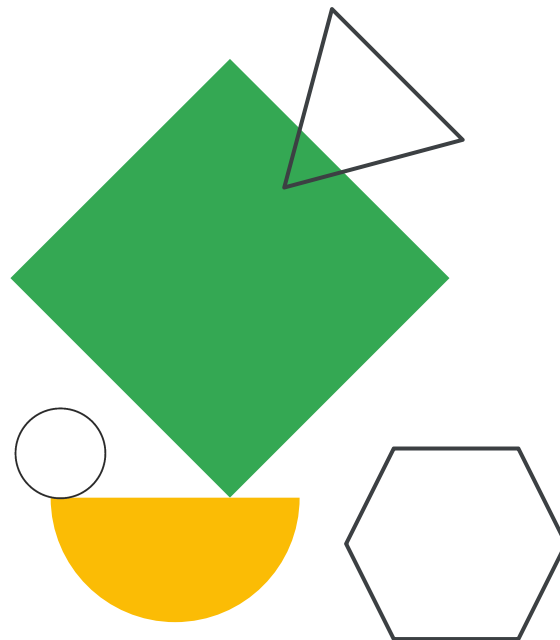
Break down costs by project and user.

Be proactive about tracking expensive queries and optimizing them.

Demo

Querying exported logs with a
dashboard

Using Looker Studio





INFORMATION_SCHEMA

Use INFORMATION_SCHEMA for monitoring

- Leverage [INFORMATION_SCHEMA views](#) (real-time updates).
- Run ad hoc SQL queries against all BigQuery jobs across your project:

```
SELECT job_id, user_email, total_bytes_processed
FROM `region-us`.INFORMATION_SCHEMA.JOBS_BY_PROJECT
WHERE EXTRACT(DATE FROM creation_time) = CURRENT_DATE()
ORDER BY total_bytes_processed DESC
LIMIT 3
```

Query complete (1.2 sec elapsed, 53 MB processed)

Job information Results JSON Execution details

Row	job_id	user_email	total_bytes_processed
1	script_job_3da815dcf7e6495a2b000dbebe6ee6c9_0	bigquery-adminbot@system.gserviceaccount.com	21906600899080
2	materialized_view_refresh_Gllh81T2G0mKELKVJyUHTSDy4JrCK	bigquery-adminbot@system.gserviceaccount.com	21906600899080
3	bqjob_r69c7484a9bf14acd_000001774768dced_1	vincegonzalez@google.com	15244347073472



INFORMATION_SCHEMA views

Dataset Metadata

SCHEMATA
SCHEMATA_OPTIONS

Table and View Metadata

TABLES
TABLE_OPTIONS
VIEWS
COLUMNS
COLUMN_FIELD_PATHS
PARTITIONS

[Full list](#)

Job Metadata

JOBS_BY_USER
JOBS_BY_PROJECT
JOBS_BY_FOLDER
JOBS_BY_ORGANIZATION

Job Timeline Metadata

JOBS_TIMELINE_BY_USER
JOBS_TIMELINE_BY_PROJECT
JOBS_TIMELINE_BY_FOLDER
JOBS_TIMELINE_BY_ORG...

Access Control

OBJECT_PRIVILEGES

Reservation Metadata

RESERVATION_CHANGES_BY_PROJECT
CAPACITY_COMMITMENT_CHANGES_BY_PROJECT
ASSIGNMENT_CHANGES_BY_PROJECT

Storage Metadata

STREAMING_TIMELINE_BY_PROJECT
STREAMING_TIMELINE_BY_FOLDER
STREAMING_TIMELINE_BY_ORG

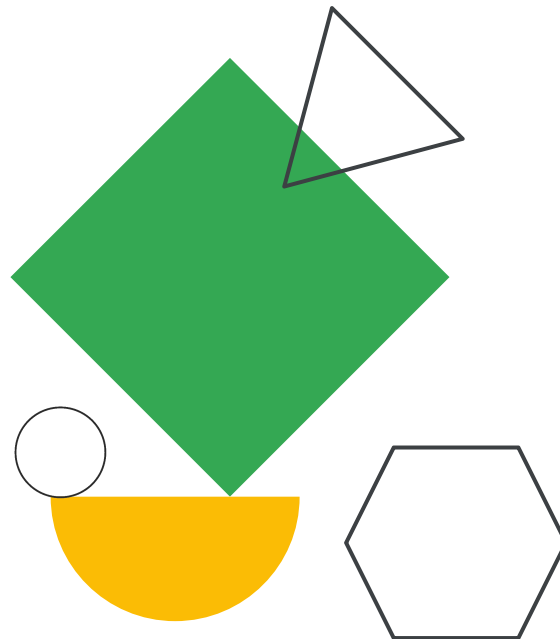
Routines Metadata

ROUTINES
ROUTINE_OPTIONS
PARAMETERS

Demo

Determining slot utilization per
project and user

Using SQL and
INFORMATION_SCHEMA



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05

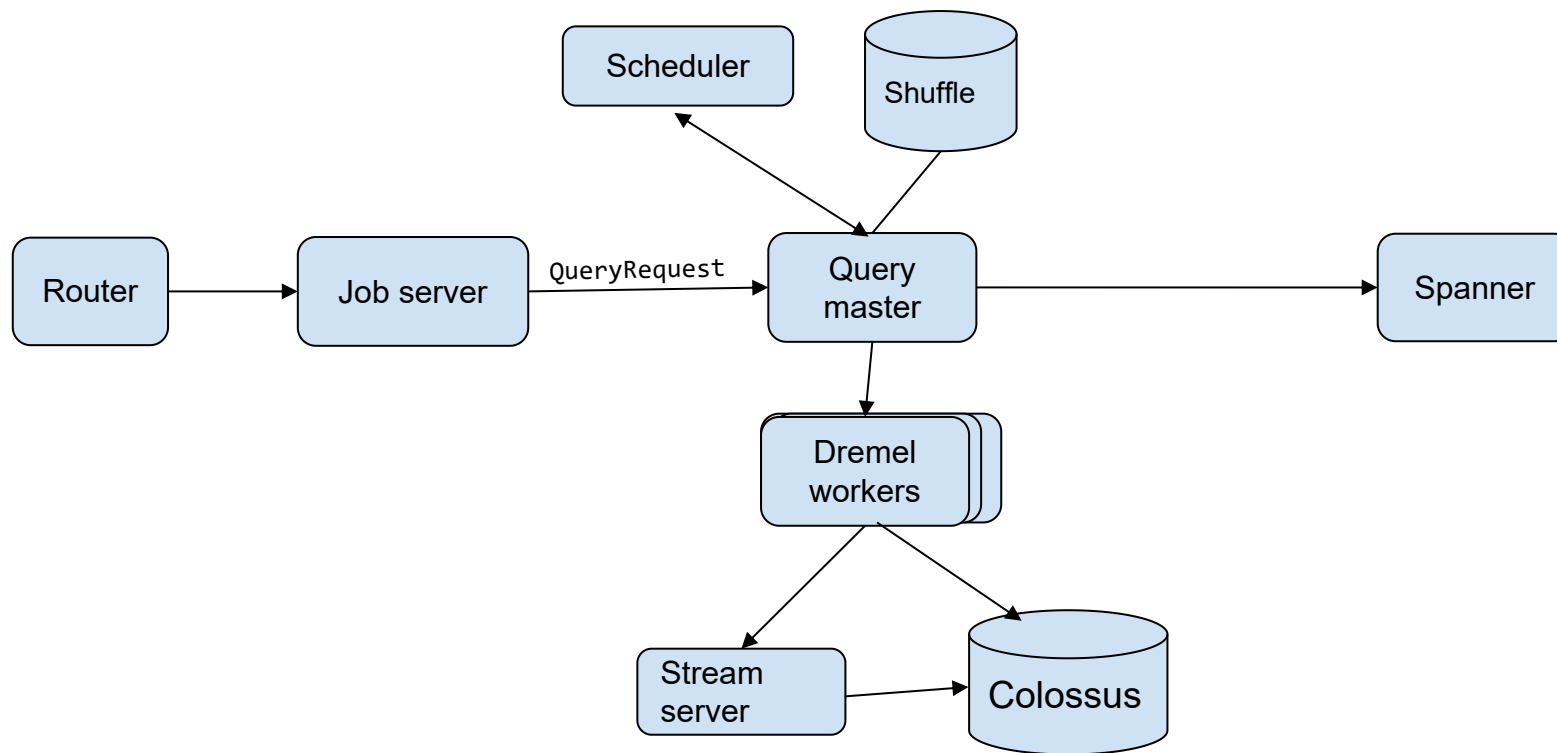
Query Path and Common Errors

Submitting a query starts with an API call

Jobs.Insert (asynchronous) / **Jobs.query** (semi-asynchronous, timeout 10s)

```
Job {  
  configuration {  
    query {  
      query: "DELETE T WHERE field = 1"  
    }  
  }  
}
```

Query path



Query startup on Job server

1. Verify user permissions to authorize job creation.
2. Extract a list of referenced entities from the SQL.
3. Authorize input tables and a destination table.
4. Read table schemas; resolve views transitively.
5. Extract referenced fields.
6. Perform billing quota checks.
7. Persist job.
8. Perform concurrency quota check.
9. Send QueryRequest to Dremel with the table schemas and URIs.

Query startup errors

1. Authorize that the job can be created.
 - Permission denied, `bigquery.jobs.create` IAM permission
2. Extract a list of referenced entities from the SQL.
 - Parses SQL, generally reports specific syntax errors
3. Authorize input tables and a destination table.
 - Permission denied on tables
4. Read table schemas; resolve views transitively.
 - Logical view referencing deleted table
5. Extract referenced fields.
6. Perform billing quota checks.
 - Insufficient quota

Query startup errors

7. Persist job.
 - Rare, massive job metadata, i.e., millions of load URLs
8. Perform concurrency quota check.
 - Usually insufficient user quota
9. Send QueryRequest to Dremel with the table schemas and URIs.
 - General query execution errors
 - “Dremel query cancelled” due to an update or maintenance.

If general Google inter-service backend issue:

- Connection error

Timeout errors have two main causes



Query is malformed or inefficient.



Query being executed has several `WITH` statements.



> 6 hours

“Results too large” error occurs if the response is larger than 10 GB compressed



Use filters to limit the result set.



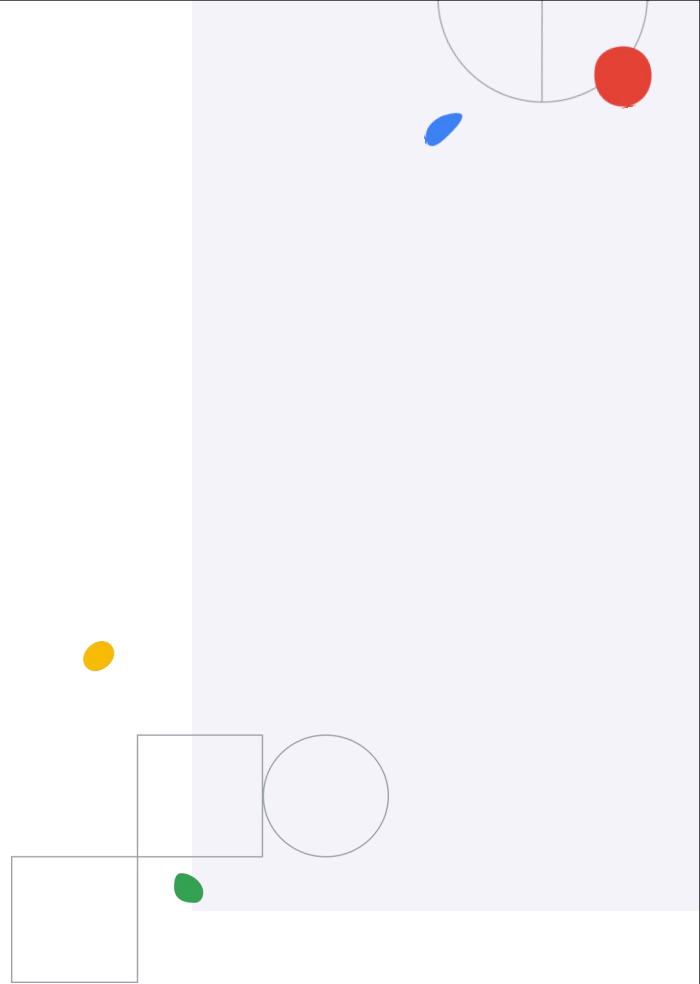
Use the LIMIT clause.



Specify a destination table for the query results.



Questions?



Lab (60 min)

Cloud Logging and Monitoring for BigQuery

1:00:00